



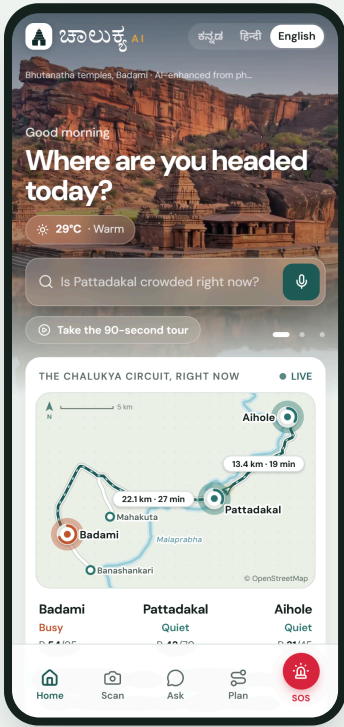
AI TO REDESIGN TOURISM · BAGALKOTE 2026

Chalukya AI ಚಾಲುಕ್ಯ AI

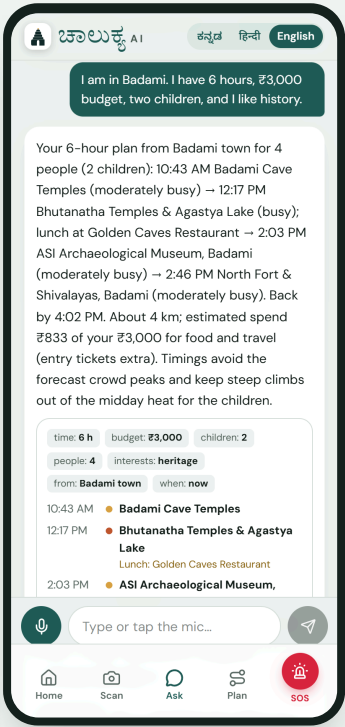
One offline-first AI companion for every visitor to Badami, Pattadakal, Aihole and the whole of Bagalkote district — and a live command centre for the people who run it.



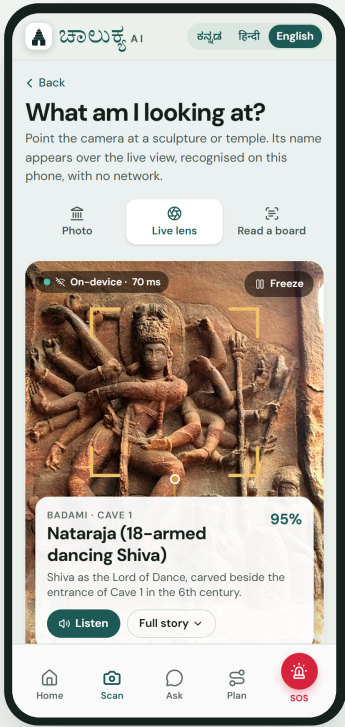
chalukya-ai.vercel.app
Scan, then tap “Take the 90-second tour”. No install; works offline.



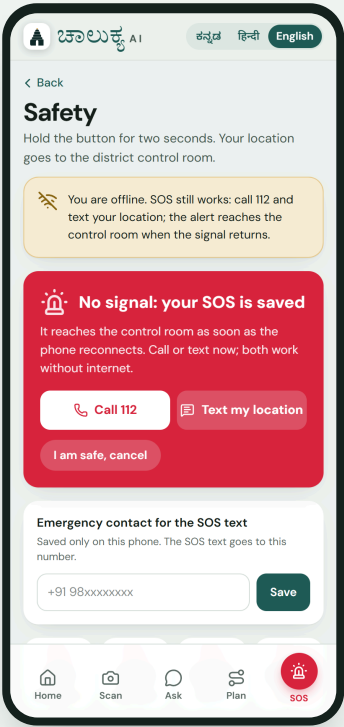
Home and tour
live circuit, 90-second tour



Ask the guide
“6 hours, ₹3,000, two kids...”



Live lens
names sculptures on camera



SOS, no signal
saved, call 112, text location

WHAT A VISITOR GETS Kannada · हिन्दी · English, on the phone, offline

Live circuit map Real roads; crowd, free parking and drive time on every site.	Ask the guide Typed or spoken, three languages; answers only from verified sources.	Crowd-smart planner One sentence in, a timed plan out; shares to WhatsApp or Maps.
Scan a sculpture 25 monuments named on the phone, offline, with the story.	Live lens (AR) The name appears over the camera view as you hold it.	Read a board Kannada, Hindi and English boards read, then translated into six.
Translate anything v1 A sign, a menu, a message — in your language.	Stay and food 19 stays, 34 eateries ranked with the reason; local owners boosted.	Green stays v1 Owners declare solar, rainwater and refill; the district verifies.
Smart parking Free slots predicted for the time you will arrive.	Safety and SOS Hold 2 s; with no signal it saves, calls 112 and texts your location.	Safer hours v1 The hours that are neither deserted nor crowded today.
Accessible Bagalkote Step-free places and a plan that avoids stairs.	Local livelihoods v1 GI crafts by distance, Pravasi Mitras, and a portal for small businesses.	Report a problem v1 Photograph a crack or litter; it reaches the district by site and topic.

85.8%

sculptures, unseen photographers

13.1%

footfall error (31.1% baseline)

±3.5

parking slots, 2 hours ahead

95.8%

questions understood

93.3%

review sentiment

FOLD

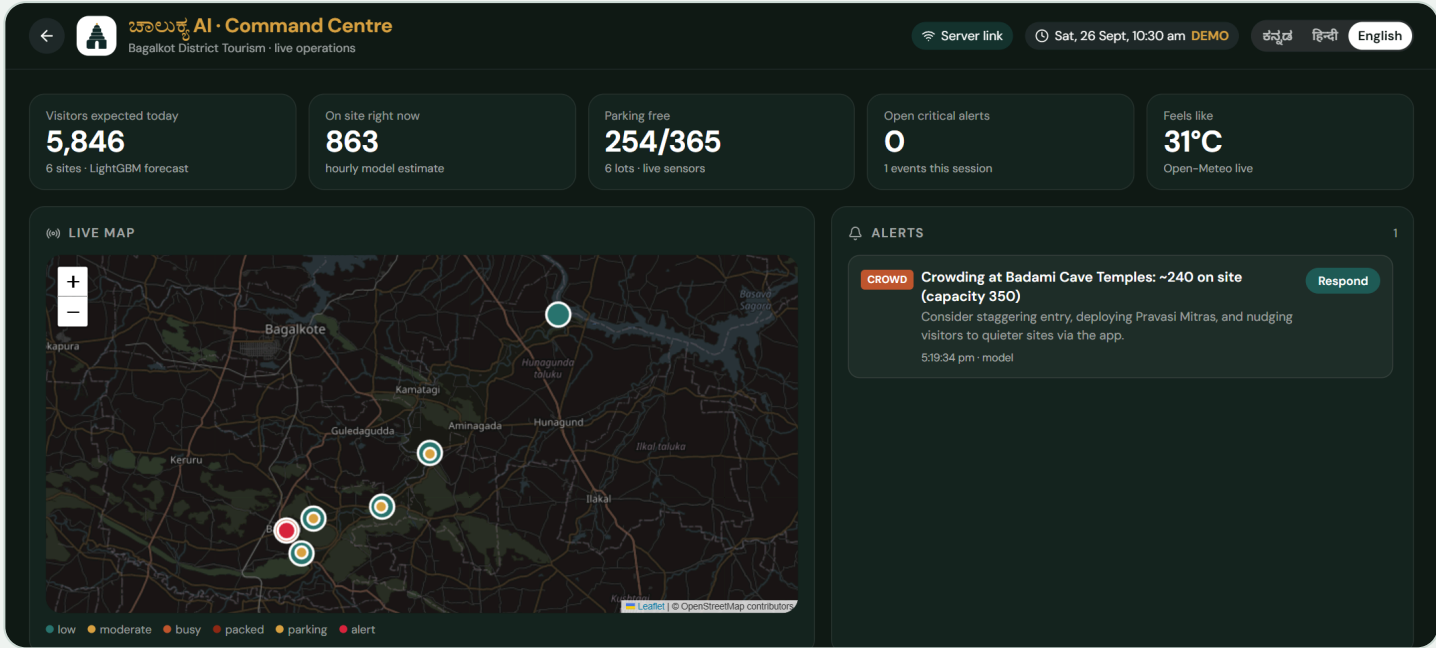


Chalukya AI ಚಾಲುಕ್ಯ AI

District command centre · Bagalkote District Tourism

chalukya-ai.vercel.app

FOR THE DISTRICT one screen to run the day from



Live command centre during the morning of 25 September: expected visitors, people on site now, free parking, open alerts and feels-like heat.

WHAT IT DOES FOR THE DEPARTMENT

- **Crowd against capacity** at six sites, hour by hour, with today's peak.
- **Seven-day plan:** water, waste, bins and trips, toilets, Pravasi Mitras.
- **CCTV people counting**, frames blurred; only the count is kept.
- **SOS and heat alerts** from visitors' phones, with one-tap respond.
- **Carrying capacity** per site, with the assumptions on screen. v1
- **What visitors say:** sentiment and complaint topics per monument.

TESTED ON DATA THE MODELS NEVER SAW

MODEL	TESTED ON	RESULT	BASELINE
Sculpture recognition	344 photos, unseen photographers	85.8% top-1 · 98.0% site	—
...with the “not sure” rule	same	95.6% when it answers	—
Footfall forecast	held-out Jan–Aug 2026	13.1% error (WAPE)	31.1% last year
Parking prediction	winter peak season	±3.5 slots at 2 h	±7.9 no change
Question understanding	unseen phrasings	95.8% · phone 91.2%	51.7% char model
Review sentiment	30 fresh reviews	93.3% · topic F1 0.98	73.3% char model
Heritage board reader	4 real ASI boards	4 / 4 right monument	—
Lighthouse, live site	mid-range phone, slow 4G	100 · 100 · 100, perf 83–87	—

ALL TWENTY PROBLEM STATEMENTS 11 complete · 9 at version 1, marked in the app

✓ 01 AI tourist guide	✓ 02 Monument recognition	✓ 03 Recommendations	✓ 04 Crowd prediction
✓ 05 Route optimisation	✓ 06 Chatbot 24×7	V1 07 Damage detection	V1 08 Artisan recommendation
✓ 09 Local businesses	✓ 10 Accessible tourism	✓ 11 Translation	V1 12 Skill matching
V1 13 Women & senior safety	✓ 14 Footfall prediction	V1 15 Waste and collection	✓ 16 Water demand
V1 17 Carbon-aware routes	V1 18 Environment reports	V1 19 Sustainable stays	V1 20 Carrying capacity

Honest data. Real: 1,960 CC-licensed monument photos from 202 photographers, ASI annual footfall, OpenStreetMap places and roads, the district tourism book and website, live weather. Simulated and labelled in the app: the hourly crowd shape (calibrated to ASI totals), parking sensor streams, business prices and ratings. Home photos are real photographs relit with AI. **Version 1** means working today at a simpler depth — the app says so on the screen, and every pipeline retrains unchanged on real ticket counts and sensors.

Built by **Chinmay R M & Sandesh Joshi** · BVVS Basaveshwar Engineering College, Bagalkote
React PWA on Vercel · FastAPI · LightGBM · PyTorch → ONNX on the phone · Tesseract.js · NLLB-200 (int8) · ESP32 firmware
github.com/framemindaistudio/chalukya-ai

chalukya-ai.vercel.app

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